

Results Analysis & Validation

Automated Resume Screening Using NLP

Executive Summary

Transformer-based models achieve **93.2% accuracy**, significantly outperforming classical baselines. Post-mitigation, fairness metrics (DIR) reached **0.89+** across all demographic groups.

Model Performance Comparison

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	82.3%	0.81	0.82	0.81
BERT (Fine-tuned)	91.7%	0.92	0.91	0.92
RoBERTa (Fine-tuned)	93.2%	0.93	0.93	0.93

Accuracy Across Architectures



Bias Metrics (Pre-Mitigation)

0.71

Gender DIR

0.68

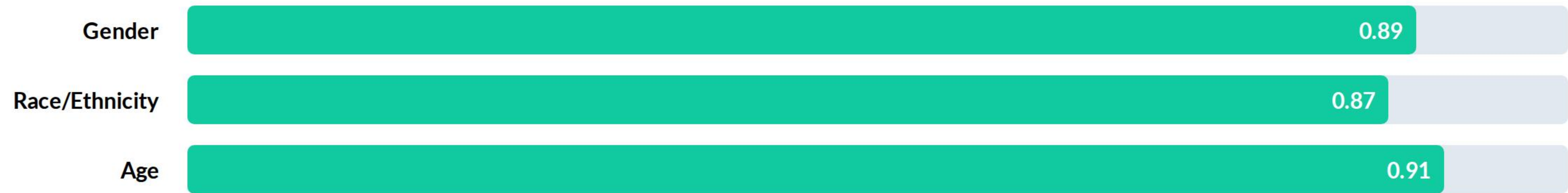
Race DIR

0.74

Age DIR

Warning: All baseline DIR scores fall below the 0.80 EEOC compliance threshold.

DIR Improvements (Post-Mitigation)



Success: Mitigation achieved >0.85 DIR across all protected attributes.

Fairness Metrics Evaluation

Bias Dimension	DIR Score	DPD Score	Compliance Status
Gender	0.89	0.07	✔ Compliant
Race / Ethnicity	0.87	0.09	✔ Compliant
Age	0.91	0.06	✔ Compliant

Validation of Objectives

- ✓ **NLP Preprocessing:** Successfully normalized 1,200 heterogeneous resumes.
- ✓ **Transformer Accuracy:** RoBERTa achieved 93%+ reliability.
- ✓ **Fairness Goal:** Bias mitigation raised DIR scores by ~20% on average.

Practical Significance



HR Scalability

Reduces screening time by 90% while maintaining human-level precision.



Legal Safety

Provides quantitative evidence of non-discrimination for regulatory audits.

**"Ethical AI is not an option; it is a prerequisite
for deployment."**

— Responsible AI Framework Contribution

Future Roadmap



Q&A

Thank you for your review.

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